Milestone 4 Progress Report: Team RAGTAG

Course: Macro Project (3rd Semester, CSE) | **Project:** Offline Privacy-First WhatsApp AI Command Center
Date: October 2026 | **Status:** All Post-Database Implementation Items Complete & Tested

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1. Project Overview & Milestone Objective

Following the initial database schema setup, Milestone 4 covers the complete data pipeline implementation, including native WhatsApp Desktop archive ingestion, multi-turn conversational thread resolution, strict priority rules, direct Google Calendar integration, purging simulation mockups in favor of real ground-truth data, dual-engine task extraction, and a 4-tier automated test suite.

2. WhatsApp Desktop ZIP Ingestion & Group Separation

- **Native ZIP & TXT Ingestion:** Added full support for WhatsApp Desktop .zip exports alongside mobile .txt files (POST /api/upload in backend/ingest.py). Decompression and validation are performed in-memory before loading messages into SQLite.
- **Automatic Group Naming:** Fixed the previous issue where all imported chats got merged into a single generic chat. The system now automatically takes the group name from the ZIP file's name (e.g., 'Aerospace Members 2026.zip' is imported under the group name 'Aerospace Members 2026').
- **Idempotent Deduplication:** Chat archives can be re-imported without creating duplicate rows or wasting storage. Messages are hashed (SHA-256) based on timestamp, sender, and content so re-imports only append new messages.
- **Real-Time Upload Loading Bar:** Added a live progress bar on the Sync page with byte-level progress (0% to 100%) and a 4-stage pipeline stepper: (1) Upload ZIP -> (2) SQLite Ingest -> (3) Extract Tasks -> (4) ChromaDB Vectors.

3. Conversational Thread Resolution & 'Is It Me?' Intelligence

In group chats, members rarely repeat @Name in every reply. The system resolves task assignees using multi-turn conversational context:

- **Above & Below Context Window:** The engine inspects a 20-minute preceding and 5-minute succeeding dialogue window. For example, when Mahima sent 'Okay then send me the report ASAP!!!', it checked who spoke right before in the exchange (Lakshyyy) and correctly assigned the task to Lakshyyy instead of Guru.
- **Same-Sender Burst Inheritance:** If a sender addresses someone in one message (e.g., Shreya saying '@~Tanishk attendance??') and then sends follow-ups ('Write branch and year also.. and check spellings'), the follow-up tasks inherit Tanishk as the assignee.
- **1st-Person Pronoun Rule:** Strict grammar parsing enforces that 'me', 'my', or 'send to me' refers to the sender of that message (e.g., Mahima or Shreya), never to Guru unless Guru sent it.
- **Pushname Handling:** Cleans WhatsApp @~Name pushnames so names match properly (e.g., @~Lakshyyy becomes Lakshyyy).
- **The 'Is It Me?' Evaluation:** Normalizes names, strips emojis, and checks against settings.json (user_name: Guru). Tasks for other group members stay off Guru's personal board.

4. Task Prioritization Rules

Replaced arbitrary priority assignments with deterministic rules based on keywords and deadlines:

Priority	Rules and Criteria
High Priority	Urgent keywords (asap, urgent, without fail, mandatory), deadlines within <= 48 hours, or directives from professors/evaluators.
Medium Priority	Regular deadlines within <= 7 days, or standard project deliverables.
Low Priority	Open-ended tasks or reading materials without a hard deadline.

5. Interactive WhatsApp-Like Workspace

- **2-Pane WhatsApp Console:** Built a WhatsApp Web-style interface with a group/contact list on the left (with unread count badges and latest snippets) and conversation stream / AI briefings on the right.
- **Direct Context Landing ('Open in Agentic Panel'):** Added 'View Context' on Task Cards and 'Open in Agentic Panel' in modals. Clicking it automatically jumps straight to that chat, scrolls to the message, and highlights it with a pulsing amber ring and badge so you can read the surrounding conversation.

6. Direct Google Calendar Integration

Replaced the manual .ics file download with a direct 'APPROVE + REMIND' button. It automatically generates a one-click Google Calendar URL (https://calendar.google.com/calendar/render?action=TEMPLATE...) pre-filling the task title, calculated due date, reminder duration, and the verified WhatsApp chat quote.

7. Purged 'Vibe-Coded' Simulation Mockups

Completely removed the fake faculty simulation buttons and mock cards. The system now operates 100% on real WhatsApp chat exports and authentic ground-truth data.

8. Dual-Engine Extraction & AI Briefings

- **Dual-Engine Architecture:** Uses local llama3 via Ollama for deep reasoning, with a 1.0-second timeout fallback to 100% offline deterministic heuristics.
- **Emergency Catch-Me-Up:** Summarizes 100+ unread messages into Key Decisions, Action Items, and Urgent Mentions.
- **Time-Duration Summaries:** Generates summaries across presets (2 hours, 24 hours, 7 days, or custom date ranges).
- **Semantic RAG Search:** ChromaDB vector search endpoint (POST /api/chat) returning verified chat quotes with exact citations.

9. 4-Tier Automated Test Suite (41 / 41 Tests Passing)

Created a comprehensive test harness (run_tests.py) validating the full system stack:

Tier	Scope & Target Subsystems	Count	Status
Tier 1	Unit & Schema: SQLite models, foreign keys, regex triggers, date parsing	17	17 / 17 Passed
Tier 2	Ingestion & Extractor: ZIP/TXT parsing, deduplication, dual-mode fallback	11	11 / 11 Passed
Tier 3	Flask API Contracts: Upload, task actions, summaries, settings, vector chat	11	11 / 11 Passed
Tier 4	E2E Simulation & Air-Gap: End-to-end task auditability, zero external network leakage	2	2 / 2 Passed
Total	**Complete System Test Coverage**	**41**	**41 / 41 Passed (100%)**

10. Conclusion & Next Steps

With the core data pipeline, conversational resolution, priority rules, and UI integrations complete and verified by 41 passing tests, the backend and frontend are production-ready. The next milestone will focus on packaging the application into a standalone desktop executable (via Electron or Tauri) for silent background execution without requiring a terminal or browser.